Introduction to Foliations

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Goal:

- Define and give examples of foliations.
- Introduce holonomy.

Outline

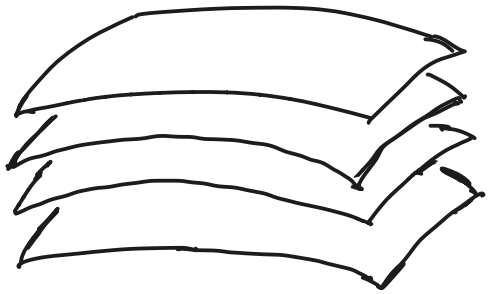
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④ Holonomy p. 10

Foliation Charts ①



A foliation of a manifold is a way to decompose it into submanifolds in a nice way. These submanifolds must all have the same dimension and should, in a sense, fit together well.

One way to formalize this is by requiring that the charts defining our manifold preserve the "leaves" of our foliation.

Definition

Let M be a smooth manifold of dimension n. A foliation atlas of codimension q is an atlas

$$\{\varphi_\alpha : U_\alpha \to \mathbb{R}^{n-q} \times \mathbb{R}^q\}$$

of M such that the transition diffeomorphisms

$$\varphi_{\alpha\beta} := \varphi_\alpha \circ (\varphi_\beta|_{U_\alpha \cap U_\beta})^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \to \varphi_\alpha(U_\alpha \cap U_\beta)$$

are of the form

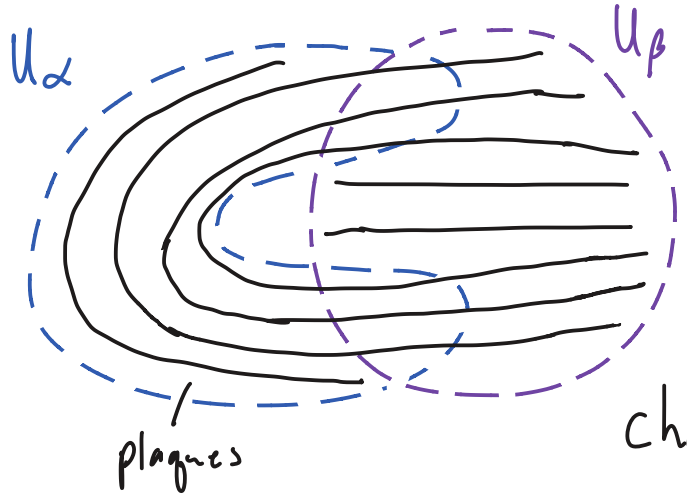
$$\varphi_{\alpha\beta}(x, y) = (g_{\alpha\beta}(x, y), h_{\alpha\beta}(y)).$$

The φ_α are called foliation charts.

The idea is that each U_α is divided into "plaques". A plaque of U_α is a connected component of

$$\varphi_\alpha^{-1}(\mathbb{R}^{n-q} \times \{y\}).$$

Our condition on the transition maps amounts to saying they map plaques to plaques.



The plaques paste together into leaves. Formally, define an equivalence relation on M by declaring $p \sim p'$ if there is a sequence of foliation charts $U_1, \ldots, U_k$ and points $p = p_0, p_1, \ldots, p_k = p'$ such that p_{j-1}, p_j lie on the

same plaque in U_j. The leaves are the equivalence classes of $\sim$.

We note that each leaf is a smooth $(n-q)$-manifold which is injectively immersed in M.

Definition

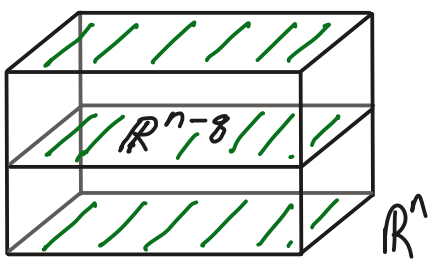
A <u>foliation</u> of codimension q of M is a maximal foliation atlas of codimension q of M. Given a foliation $\mathcal{F}$ of M, the pair $(M, \mathcal{F})$ is called a <u>foliated</u> manifold. The <u>space of leaves</u> is the quotient space $M/\mathcal{F}$.

Any foliation atlas $\mathcal{A}$ is contained in a unique maximal one given by taking all charts compatible as foliation charts to the charts in $\mathcal{A}$.

Moreover, we can show two foliation atlases define the same foliation if and only if they induce the same partition of M into leaves.

A map of foliated manifolds $f: (M, \mathcal{F}) \to (M', \mathcal{F}')$ is a map $f: M \to M'$ that maps the leaves of $\mathcal{F}$ into the leaves of $\mathcal{F}'$.

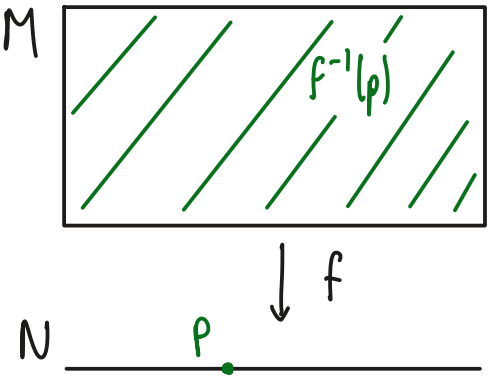
Examples ①



(1) $\mathbb{R}^n$ admits the trivial codimension q foliation given by the foliation atlas

$$\{ \mathrm{id} : \mathbb{R}^n \longrightarrow \mathbb{R}^{n-q} \times \mathbb{R}^q \}.$$

(2) More generally, any submersion $f: M \to N$ defines a foliation $\mathcal{F}(f)$ whose leaves are the connected components of the fibers of f.



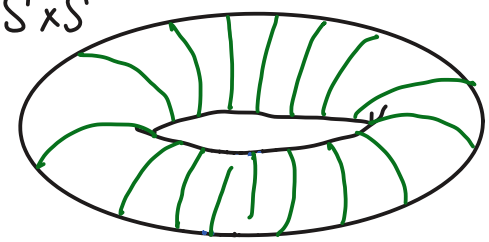
A foliation atlas is obtained from charts where f looks locally like $\mathbb{R}^n \to \mathbb{R}^{n-q}$.

Such foliations are called <u>simple</u>. If all fibers of f are connected, we call it <u>strictly simple</u>.

(3) Consider the submersion $s: \mathbb{R}^2 \to \mathbb{R}$ given by

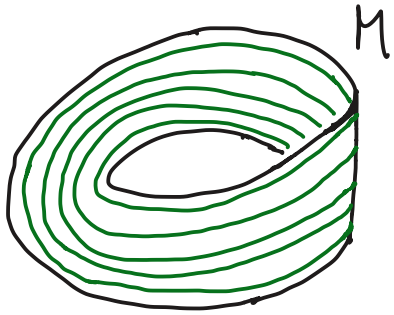
$$s(x,y) = x - y\sqrt{2}.$$

We get a codimension 1 foliation $\mathcal{F}(s)$ of $\mathbb{R}^2$. Via the covering map $p: \mathbb{R}^2 \to S^1 \times S^1$, we get a foliation of $S^1 \times S^1$.



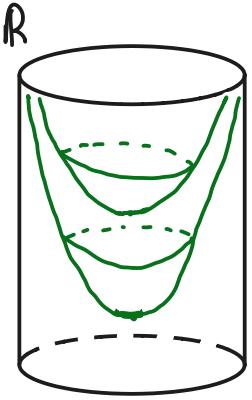
The leaves are diffeomorphic to $\mathbb{R}$ and are dense in $S^1 \times S^1$.
This is the <u>Kronecker foliation</u>.

(4) Via the covering map $p: \mathbb{R}^2 \to M$, we get a codimension 1 foliation of the Möbius strip from the trivial codimension 1 foliation of $\mathbb{R}^2$. All the leaves are diffeomorphic to S^1 and wrap around M "twice" except for the "middle" S^1.



If M is a manifold with boundary, we can still define foliations. Usually, we require either that the leaves are transverse to the boundary, or that the components of the

boundary are leaves as well.

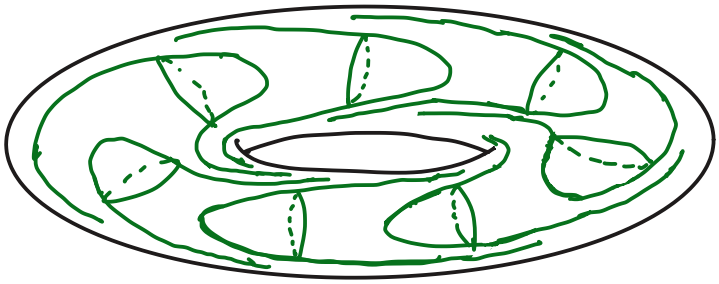


(5) Consider the submersion $s: \mathbb{D}\times\mathbb{R} \to \mathbb{R}$ given by

$$s(z,x) = \exp\left(\frac{1}{1-|z|^2}\right) - x.$$

We get a codimension 1 foliation $\mathcal{F}(s)$ of $\mathbb{D}\times\mathbb{R}$. Via the covering map $p: \mathbb{D}\times\mathbb{R} \to \mathbb{D}\times S^1$, we get a foliation of $\mathbb{D}\times S^1$. Adding the boundary torus as a leaf, we get a foliation $\mathcal{R}$ of $\mathbb{D}\times S^1$. All other leafs are diffeomorphic to $\mathbb{R}^2$.

Reeb foliation



(6) The sphere S^3 can be viewed as two solid tori glued along their boundary. Since the boundary is a leaf of the Reeb foliation, it follows we can glue the foliations to get the Reeb foliation of S^3.

Alternative Definitions ②

The data of a codimension q foliation can be expressed in many alternative ways. For example, taking the tangent spaces to each leaf in the foliation defines a subbundle of the tangent bundle. Conversely, given a subbundle of TM with some conditions, we will be able to "integrate" the subbundle to a foliation of M.

Definition

A subbundle E of TM is called <u>involutive</u> if E is closed under the Lie bracket. That is, if $X, Y \in \Gamma(E)$, then $[X,Y]$ is also a section of E.

Theorem (Frobenius Integrability)

Let E be a rank $n-q$ involutive subbundle of TM. For any point $x \in M$, there exists an open neighborhood $U \subseteq M$ and a diffeomorphism $\varphi : U \to \mathbb{R}^{n-q} \times \mathbb{R}^q$ such that $E|_U = \ker(d(pr_2 \circ \varphi))$.

Proof

We may pick a neighborhood $\tilde{U}$ of x and a smooth map $\psi : U \to \mathbb{R}^{n-q}$ such that $d\psi$ is injective on $E|_{\tilde{U}}$.

Define a frame $V_1, \dots, V_{n-q}$ on E such that $d\psi^i(V_j) = \delta^i_j$.

Since E is involutive, there are functions C^k_{ij} such that

$$[V_i, V_j] = \sum_{K=1}^{n-q} C^k_{ij} V_K .$$

Then, $C^k_{ij} = d\psi^k([V_i, V_j]) = V_i \cdot d\psi^k(V_j) - V_j \cdot d\psi^k(V_i) = 0$, so the V_j commute. Let ρ_j^t denote the time t flow of V_j.

We may pick a neighborhood V of 0 in $\mathbb{R}^q$ and a smooth $\phi : V \to M$ such that $\phi(0) = x$ and

$$E_x \oplus d\phi_0(\mathbb{R}^q) = T_x M .$$

Define $\tilde{\varphi}$ from a neighborhood of 0 in $\mathbb{R}^{n-q} \times \mathbb{R}^q$ to M by

$$\tilde{\varphi}(t_1, \dots, t_{n-q}, p) = \rho_1^{t_1} \circ \dots \circ \rho_{n-q}^{t_{n-q}} \circ \phi(p) .$$

Then, $d\tilde{\varphi}_{(0,0)}(\tau_1,\dots,\tau_{n-q},\eta_1,\dots,\eta_q) = \tau_1 v_1 + \dots + \tau_{n-q} v_{n-q} + d\phi_0(\eta_1,\dots,\eta_q)$,

so $d\tilde{\varphi}_{(0,0)}(\mathbb{R}^{n-q}\times\mathbb{R}^q) = T_x M$. Thus, $\tilde{\varphi}$ is a local diffeomorphism at $(0,0)$. Take $\varphi: U \to \mathbb{R}^{n-q}\times\mathbb{R}^q$ to be the inverse of $\tilde{\varphi}$ in a sufficiently small neighborhood of x. Then,

$$d\varphi(E|_u) = \mathbb{R}^{n-q}\times\{0\},$$

so that $E|_u = \ker(d(p_2\circ\varphi))$, as desired. ■

Proposition

The following data are equivalent:

(i) A foliation $\mathcal{F}$ of M of codimension q.

(ii) An involutive subbundle E of TM of rank $n-q$.

(iii) An open cover $\{U_\alpha\}$ of M with submersions $s_\alpha: U_\alpha \to \mathbb{R}^q$ such that there are diffeomorphisms

$$\gamma_{\alpha\beta}: s_\beta(U_\alpha\cap U_\beta) \longrightarrow s_\alpha(U_\alpha\cap U_\beta)$$

with $\gamma_{\alpha\beta}\circ s_\beta|_{U_\alpha\cap U_\beta} = s_\alpha|_{U_\alpha\cap U_\beta}$.

Haefliger cocycle

(iv) A locally trivial differential graded ideal $\mathcal{I}$ of rank q in the DGA $\Omega(M)$.

Proof

There is a lot to check, but the hardest implication is given by the Frobenius integrability theorem. □

Remarks

- Any line field is involutive, so it defines a codimension $n-1$ foliation.

- If ω is a nowhere-vanishing 1-form on M, it defines a codimension 1 foliation if and only if $d\omega \in \Omega(M) \wedge \omega$. That is,

$$d\omega \wedge \omega = 0.$$

- If $\omega = df$ for f a submersion, we recover $\mathcal{F}(f)$ from above.
- The bundle E above is called the tangent bundle of $\mathcal{F}$ and denoted $T\mathcal{F}$.
- A smooth map $f: M \to M'$ is a map of foliated manifolds $(M, \mathcal{F}) \to (M', \mathcal{F}')$ if and only if $df(T\mathcal{F}) \subset T\mathcal{F}'$.

Definition

We say that $\mathcal{F}$ is orientable if $T\mathcal{F}$ is orientable.

We say that $\mathcal{F}$ is transversely orientable if $T\mathcal{F}$ is co-orientable.

If $\mathcal{F}$ is transversely orientable, it is given by a nowhere-vanishing one-form ω on M. Then, there is some one-form α such that

$$d\omega = \alpha \wedge \omega$$

It turns out that

$$gv(\mathcal{F}) = [\alpha \wedge d\alpha] \in H^3_{dR}(M)$$

is independent of the choice of ω and α, and thus gives an invariant of the foliated manifold.

Making more foliations

Products: If $(M, \mathcal{F})$ and (N, G) are two foliated manifolds,

We get the product foliation $(M \times N, \mathcal{F} \times \mathcal{G})$ given by

$$T(\mathcal{F} \times \mathcal{G}) = T\mathcal{F} \oplus T\mathcal{G}.$$

Pull-backs: Let $f: N \to M$ be smooth, $\mathcal{F}$ a foliation of M of codimension q. We say f is transversal to $\mathcal{F}$ and write $f \pitchfork \mathcal{F}$ if for all $x \in N$,

$$df_x(T_xN) + T_{f(x)}\mathcal{F} = T_{f(x)}M.$$

If $f \pitchfork \mathcal{F}$ we get the pull-back foliation $f^*\mathcal{F}$ given by

$$T(f^*\mathcal{F}) = df^{-1}(T\mathcal{F}).$$

Quotients: Let $(M, \mathcal{F})$ be foliated and let G be a group with a free, properly discontinuous action by diffeomorphisms on M. Then, the quotient manifold M/G is Hausdorff. Assume $\mathcal{F}$ is G-invariant, so that $dg(T\mathcal{F}) = T\mathcal{F}$ for all $g \in G$. We get the quotient foliation $\mathcal{F}/G$ given by

$$T(\mathcal{F}/G) = dp(T\mathcal{F}),$$

where $p: M \twoheadrightarrow M/G$ is the quotient map.

Flat Bundles: Let $G = \pi_1(M)$ and let $\tilde{M}$ be the universal cover of M. Recall that G acts freely and properly discontinuously on $\tilde{M}$, with $\tilde{M}/G = M$. We will write this as a right action. Suppose G acts on the left on some manifold F. We get the space

$$E = \tilde{M} \times_G F,$$

which in particular is a manifold.

We get a foliation on E by taking $\mathcal{F} = \mathcal{F}(\pi : \tilde{M} \times F \to F)/G$.

Holonomy

Consider the germs of diffeomorphisms $(M, x) \to (M, x)$.

Fact: The set of such germs forms a group, which we denote

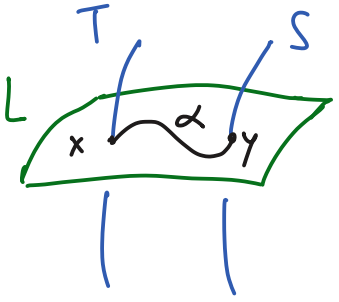
$$\mathrm{Diff}_x(M).$$

Let $(M, \mathcal{F})$ be foliated, L be a leaf, $x, y \in L$ and T, S be transversal submanifolds to $\mathcal{F}$ at x, y respectively. To any path α from x to y we will associate a germ

$$hol(\alpha) = hol^{S,T}(\alpha) : (T, x) \to (S, y).$$

If there is a foliation chart U such that $\alpha([0,1]) \subset U$, then take any $f : A \subset T \to S$ that is plaque preserving and maps $x \mapsto y$. We define $hol^{S,T}(\alpha)$ as the germ of f.

If α is too big, decompose it into small enough paths and define $hol^{S,T}(\alpha)$ as the composition of all the small holonomies.



We can check that for a transversal submanifold T at $x \in L$, this defines a homomorphism

$$hol^T = hol^{T,T} : \pi_1(L, x) \to \mathrm{Diff}_x(T).$$

Note that $\mathrm{Diff}_x(T) \cong \mathrm{Diff}_0(\mathbb{R}^q)$, so we obtain a homomorphism

$$hol : \pi_1(L) \to \mathrm{Diff}_0(\mathbb{R}^q).$$

Its image $\mathrm{Hol}(L)$ is well-defined up to conjugation and is called the holonomy group of L.